TECHNICAL ENTITLEMENT GUIDE

# SpeedCloud

Customer entitlements and operating responsibilities

A practical explanation of the managed services, support boundaries, SLAs, and customer actions described in the supplied entitlement document.




Source document cover reproduced for context. Service scope and commercial entitlements remain subject to the applicable agreement.

01 | ORIENTATION

# How to read the entitlement model

The supplied document describes SpeedCloud as a sovereign, enterprise-oriented cloud built on open-source infrastructure and backed by managed services. An entitlement is best understood as a service boundary: what NxtGen operates, what the customer must provide or approve, and what is excluded or chargeable.

| Layer | Documented foundation | Technical question |
|---|---|---|
| **Cloud platform** | Red Hat Enterprise Linux 9, OpenStack 17.1, OpenStack APIs, and Red Hat Ceph Storage for block, file, and object storage. | Which compute, network, storage, and API capabilities are included in the subscribed environment? |
| **Infrastructure** | CPU, memory, storage, high-speed networking, and GPU support, with a growth-ready architecture. | What performance, quota, placement, and scaling limits apply to the workload? |
| **Managed operations** | VMs, OS management, Kubernetes, backup, endpoint detection and response, firewall, monitoring, migration, and DR-as-a-Service. | Which recurring operational tasks are performed by NxtGen and which remain with the customer? |
| **Governance** | Formal requests, change management, incident handling, patch cadence, service levels, and a shared responsibility model. | Who approves changes, supplies access, responds to alerts, and validates recovery? |

## Core customer outcomes

- Sovereignty: the document emphasizes data residency and operational control within India.
- Self-service: customers can provision and manage resources through the cloud portal and APIs, subject to scope and entitlements.
- Managed operations: NxtGen can take on selected infrastructure, security, backup, and lifecycle tasks.
- Measured service: availability, response targets, maintenance windows, exclusions, and service credits define the operating contract.

Use this guide as an explanation of the supplied document, not as a replacement for the signed order form, SLA, or service-specific scope.

02 | CORE MANAGED SERVICES

# What is managed and when to use it

The document describes managed capabilities that reduce day-to-day infrastructure work. The exact items marked as available on request should be confirmed in the customer order or service schedule.

| Service | What NxtGen manages | Example |
|---|---|---|
| **VM management** | Deployment, monitoring, patching, maintenance, security measures, backups, and support for customer VMs. Isolated and routed external networks are listed as on-request scope. | Run a three-tier business application while the customer owns the application configuration and data. |
| **OS management** | Support for stated Windows and Linux families, user/group administration, access maintenance, performance tuning, and selected OS-level tasks on request. | Keep a RHEL or Windows fleet patched and access-controlled without assigning every task to the application team. |
| **Kubernetes management** | Centralized cluster lifecycle management, networking, storage, security setup, monitoring, logging, upgrades, and incident support as subscribed. | Operate a production container platform while application teams deploy services through approved namespaces and policies. |
| **Backup** | Backup agent and server setup, weekly full backup, daily system backups, job-status reporting, retention and failure notifications, and restore support within scope. | Recover a VM or application dataset after accidental deletion or infrastructure failure, then verify the recovery process. |

## Backup schedule described in the source

- System backup: 6 days per week.
- Weekly full backup: 1 day per week, with a four-week schedule shown.
- The source notes that the initial backup of a newly installed system is performed before the regular weekly cycle begins.

Technical check: confirm the actual retention period, restore granularity, application-consistency method, recovery-point objective, recovery-time objective, and any storage or source-data charging basis.

03 | SECURITY SERVICES

# Endpoint, firewall, and access controls

The entitlement document separates platform security from customer workload security. The managed services add controls, but they do not remove the customer's responsibility for identity, application security, data, and configuration choices.

| Capability | Included behavior described | Important boundary |
|---|---|---|
| **Managed EDR** | Agent deployment, baseline policies, custom indicators and rules, USB/peripheral policy, host firewall policy, dashboards, logs, and SOC support for Windows, Linux, and macOS endpoints. | Customer provides endpoint inventory and access, responds to alerts, and remains responsible for actions outside the agreed scope. |
| **Managed firewall** | Deployment, baseline configuration, HA where two firewalls are provisioned, policies, NAT, routing, IPsec and SSL VPN configuration, IPS rules, geo-filtering, backups, monitoring, and formal policy changes. | Training, custom integrations, hardware relocation, third-party issues, and items outside defined scope are excluded or require separate agreement. |
| **Remote access** | The source describes SSL VPN remote access for up to 5 users by default and up to 5 site-to-site IPsec tunnels by default; additional usage is chargeable. | Credential sharing is not allowed; client-side, internet, unsupported guest OS, and some topology issues are out of scope. |
| **Logging** | Local firewall logs and syslog forwarding to customer-managed servers are described; long-term retention and historical reporting require SOC/SIEM services. | Confirm log retention, storage, search, alerting, and evidence requirements before assuming the default service is a full SIEM. |

## Example: secure remote administration

- Use MFA and SSL VPN for administrator access.
- Restrict the VPN user list and separate administration from application traffic.
- Apply firewall and host controls, then send logs to the agreed monitoring or SIEM destination.
- Test access revocation and incident escalation before production onboarding.

The goal is a controlled access path with clear evidence of who can reach which management surface and how quickly access can be removed.

04 | VISIBILITY AND CONTINUITY

# Monitoring, migration, and disaster recovery

These services address the operating lifecycle: observe the environment, move workloads into or across environments, and recover from a major outage or data-loss event.

| Service | Technical role | Example use |
|---|---|---|
| **Monitoring and logging** | Prometheus and Grafana are named for metrics and visualization; the document also references an Elastic-based logging stack for search and analysis. | Create dashboards for VM health, capacity, latency, and service availability, then route actionable alerts to the support process. |
| **Migration as a service** | Replication-based migration across physical, virtual, private-cloud, and remote-datacenter environments, with test cutovers, secure transfer, bandwidth management, and API integration. | Move a legacy VM into SpeedCloud with a controlled cutover and a rollback plan rather than a long application outage. |
| **DR-as-a-Service** | Automated failover and failback using immutable storage and secure backup; RTO and RPO are maintained and the source says the DR plan is tested once every three months for subscribed customers. | Recover a critical application in a secondary environment, validate the runbook quarterly, and record actual recovery time. |

## Continuity questions to confirm

- What is the agreed RPO for each application and data tier?
- What is the agreed RTO, and how is it measured during a test?
- Which systems, dependencies, DNS changes, credentials, and network paths are included in failover?
- Who approves a failover, who communicates to stakeholders, and who signs off the return to primary?

A backup exists to restore data; a DR service exists to restore an operating service. The runbook, dependencies, and test evidence are what connect the two.

05 | OPERATIONS

# Support, incidents, patching, and changes

The document defines operational processes that protect stability and traceability. These processes matter as much as the infrastructure specification when the environment is business-critical.

| Process | What the source describes | Customer action |
|---|---|---|
| **Incident response** | Incidents are logged, tracked, and closed through NxtGen's service process. Severity levels cover urgent outages through low-impact requests. | Provide impact, affected services, timestamps, access, logs, and a reliable technical contact. |
| **Response targets** | The document contains service-specific targets including Critical under 30 minutes, High under 4 hours, Medium/Low under 8 hours, and a separate severity matrix of 1 business hour, 4 business hours, 1 business day, and 2 business days. | Confirm which matrix applies to the subscribed service and whether the clock is business-hours or 24x7. |
| **Patching** | Underlying infrastructure and hypervisor patches are applied periodically within agreed maintenance windows; emergency patches may be applied when necessary. OS and application changes may require customer approval. | Maintain supported images, test application compatibility, approve impact-bearing changes, and keep rollback plans. |
| **Change management** | Changes are standard, normal, or emergency. Formal requests and customer approval are used for non-standard changes, except urgent restoration activity. | Raise complete requests, identify dependencies, define a maintenance window, and validate the result after change. |

## Raise a request

- Use the NxtGen cloud portal or the support email listed in the source document.
- For urgent or business-impacting issues, use the service-desk phone contacts listed in the agreement.
- Include the environment, service, impact, severity, error evidence, and requested outcome.

Operational quality improves when a request is written so that the service desk can reproduce or route it without repeated clarification.

06 | SLA AND SERVICE CREDITS

# How availability commitments should be interpreted

The supplied document describes an infrastructure SLA with definitions of unavailability, service credits, exclusions, and customer remedies. Treat the signed SLA as the governing source for production decisions.

| Item | Source-level meaning | What to confirm |
|---|---|---|
| **Availability** | The source states a 99.5% uptime commitment for the managed service, excluding scheduled maintenance, and defines unavailability for service, storage I/O, and network-connect situations. | Which product is measured, which endpoints are tested, what monitoring evidence is accepted, and which maintenance windows apply. |
| **Service credits** | The document shows extensions of 1 day for 99.50%-99.00%, 2 days for 99.0%-98.0%, and 7 days below 98% availability. | Credit calculation, claim timing, eligible products, tenure treatment, and whether credits are the exclusive remedy. |
| **Exclusions** | Examples include non-payment, force majeure, internet or site connectivity outside NxtGen control, customer or third-party equipment, credentials, unsupported configurations, and other stated exclusions. | Map every dependency and customer-controlled component so an outage is not incorrectly assumed to be covered. |

## Important reading rule

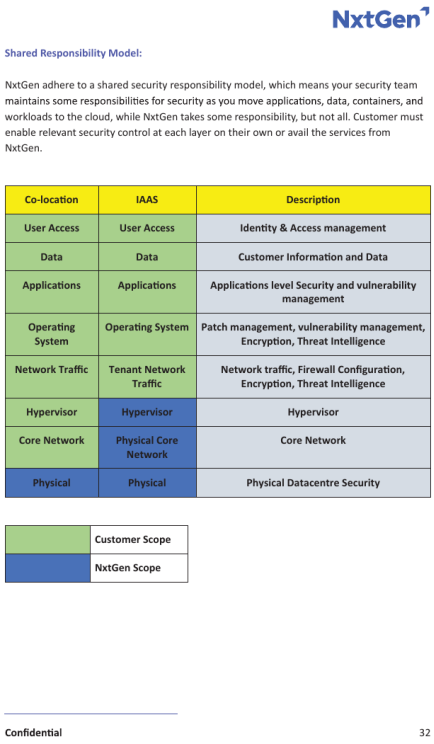
An availability SLA is not a blanket performance guarantee. The source separately states that vDC performance depends on resource sizing and does not carry the same kind of commitment. Measure availability, latency, IOPS, throughput, and application response time as separate acceptance criteria.

Before signing off a production design, build an entitlement matrix that maps each required service to its exact subscription, SLA, support hours, exclusions, customer inputs, and commercial treatment.

07 | SHARED RESPONSIBILITY

# Who owns which layer

The document explicitly uses a shared security responsibility model. NxtGen protects the managed cloud foundation, while customers remain accountable for their access, data, applications, and workload configuration.



NxtGen

Shared Responsibility Model:

NxtGen adhere to a shared security responsibility model, which means your security team maintains some responsibilities for security as you move applications, data, containers, and workloads to the cloud, while NxtGen takes some responsibility, but not all. Customer must enable relevant security control at each layer on their own or avail the services from NxtGen.

| Co-location | IAAS | Description |
|---|---|---|
| User Access | User Access | Identity & Access management |
| Data | Data | Customer Information and Data |
| Applications | Applications | Applications level Security and vulnerability management |
| Operating System | Operating System | Patch management, vulnerability management, Encryption, Threat Intelligence |
| Network Traffic | Tenant Network Traffic | Network traffic, Firewall Configuration, Encryption, Threat Intelligence |
| Hypervisor | Hypervisor | Hypervisor |
| Core Network | Physical Core Network | Core Network |
| Physical | Physical | Physical Datacentre Security |

| | |
|---|---|
| | Customer Scope |
| | NxtGen Scope |

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Figure 1. Shared responsibility visual reproduced from the supplied entitlement document.

| Layer | Customer focus | NxtGen focus |
|---|---|---|
| **User access** | Identity, access approvals, MFA choices, least privilege, and credential lifecycle. | Identity and access capabilities within the managed service boundary. |
| **Data and applications** | Customer information, application security, vulnerability management, configuration, and recovery requirements. | Infrastructure controls and optional managed security services. |
| **Operating system and network** | Workload configuration, application connectivity, and approvals for changes. | Patch management, vulnerability management, firewall configuration, encryption, and threat intelligence as subscribed. |
| **Hypervisor, core network, and physical** | Validate service dependency and provide site or access inputs where required. | Hypervisor, core network, data-center infrastructure, and physical security within the service boundary. |

08 | PERFORMANCE EVIDENCE

# How to interpret the benchmark section

The entitlement document includes comparative benchmark tables for FIO, Sysbench, Geekbench, and iPerf. These figures are useful for forming hypotheses, but they are not a substitute for a workload-specific test plan.

| Test | SpeedCloud figure shown | Technical interpretation |
|---|---|---|
| **FIO 4K read** | 785k IOPS | A storage result for a particular block size, queue depth, device, and test configuration. Validate with the customer's I/O profile. |
| **FIO 4K write** | 199k IOPS | Write behavior may differ from read behavior; validate durability, caching, queue depth, and latency under sustained load. |
| **Sysbench transfer** | 46.6k MiB transferred; 4.66k MiB/sec; 0.1 ms max latency shown | A memory or transfer-oriented result. Confirm the exact test, concurrency, memory size, and whether the result is peak or sustained. |
| **Geekbench** | 1.8k single-core; 9.7k multi-core | Use single-core results for latency-sensitive threads and multi-core results for parallel workloads; test the actual VM or bare-metal shape. |
| **iPerf** | 30.8 Gbits/sec sender; 30.7 Gbits/sec receiver; 35.9 GBytes transfer shown | Network throughput under a particular topology and concurrency. Validate application-level transfer time and east-west traffic patterns. |
| **Infrastructure claims** | Up to 6400 MT/s memory transfer and 200 Gbps internal bandwidth per host are described in the source. | Treat these as platform capability claims; confirm actual SKU, adapter, topology, quota, and tenant-level availability. |

## Recommended benchmark method

- Use a representative dataset, concurrency, packet size, storage tier, VM shape, and application version.
- Measure p50, p95, and p99 latency where relevant, not only an average or maximum.
- Record sustained throughput, job completion time, CPU/GPU utilization, and failure behavior.
- Compare complete cost, including storage, egress, backup, support, licensing, and engineering effort.

The supplied tables compare SpeedCloud with AWS, GCP, and Azure, but the correct business decision depends on the customer's workload and the assumptions behind the comparison.

09 | PRACTICAL CLOSEOUT

# Entitlement checklist before production

Use this checklist during solution design, contract review, and the production readiness review.

| Area | Questions to close |
|---|---|
| **Subscribed services** | Which VM, OS, Kubernetes, backup, EDR, firewall, monitoring, migration, and DR services are explicitly included? Which are on request or chargeable? |
| **Customer inputs** | Have the customer supplied access, endpoint inventory, network information, application owners, maintenance approvals, contacts, and recovery priorities? |
| **Security boundary** | Who owns identity, data, application vulnerabilities, OS configuration, network policies, encryption, logs, and incident decisions? |
| **Support boundary** | Which response-time matrix applies? Are support hours, severity definitions, escalation routes, OEM dependencies, and exclusions documented? |
| **Resilience** | Are backup retention, restore testing, RTO, RPO, failover, failback, and quarterly DR tests agreed and evidenced? |
| **Commercials** | Are licenses, VPN users and tunnels, log retention, peak source data, egress, GPU usage, support, service credits, and custom work priced? |
| **Acceptance** | Have availability, performance, security, monitoring, backup, restore, and change-management tests been passed before go-live? |

## Technical conclusion

The customer-entitlements document is strongest when used as an operating contract: it defines the service foundation, optional managed capabilities, shared responsibilities, support processes, and the evidence needed to evaluate availability and performance. The next step should be a workload-specific entitlement matrix and proof of concept tied to the signed scope.

This guide summarizes and explains the supplied document. Product availability, final limits, pricing, certifications, SLA terms, and service credits must be confirmed in the applicable agreement and service schedule.